## Factors to consider in a power analyser

Anuj Rohilla, deputy general manager-products, SPI Engineers Pvt Ltd, has listed out some points to be considered when going for a power analyser:

- Instrument should have high accuracy and sampling rate.
- Handheld power quality analysers are fairly lightweight (generally 2kg to 2.5kg) and measure a variety of parameters. Typical parameters include voltage, amperage, frequency, dips (sags) and swells in voltage values, power factor, harmonic currents, and resulting distortion and crest factor, power and energy, voltage and current unbalance, inrush current values and light flicker.
- Recorded data should be downloaded in an easy way to detect problems and carry out the challenging measurements.
- It should easily generate reports that can be sent from one place to another.
- For certain applications, you should be able to plug in the device to the mains directly or connect to the power supply module provided. Some other applications may need a battery that can support the device for significant periods of time.
- Some devices are handheld and others have to be placed on a surface or stand. The size and weight of power quality analyser should be considered if it is to be used frequently and for longer periods.

Apart from being used as energy-measuring devices, power analysers can also be used for network analysis and determination of harmonics and phase rotation. Incorporation of digital displays adds to the ease of use. PA900 from Vitrek delivers multichannel, wideband performance and its coloured touchscreen user interface is easy to use.

## Handling controls remotely: safety or ease

Electronic circuitry is improving day by day. New materials are under research to overcome attenuation and loss issues in electronic devices. Specs are getting better, resulting in handing over more and more control to the user. Madhukar Tripathi, senior manager - marketing and channel sales, Anritsu India Pvt Ltd, highlights some of these controls as, "user settings to control measurement speeds and noise floor."

KM2200 from Kusam-Meco allows connection through local area network for remote control and measurement. Remote connections also have a safety angle.

***Better safe than sorry.*** Safety with electrical equipment has always been an issue. Equipment often blow up, either due to shorting or overload, causing personal injury and equipment damage. The good news, however, is that these misfortunes are avoidable.

Some features are built in the system, like CAT rating requirements. Others are basic guidelines for safety. Tripathi says, "Connection of cable and antenna need special attention." These should not be too loose or tightly connected. He adds, "After use, all cables and connectors must be kept safely, and caps should be used to cover connectors. This helps to save connectors and improve their life usage."

KM2100 from Kusam-Meco features CAT III 1000V/CAT IV 600V for safety in addition to a protection shield in the multifunction harmonic and power analyser.

## Equipment capture data, while the computer processes it

On an average, T&M equipment cost a lot. These have to be calibrated at regular intervals, as well. Proper maintenance, usage and precautions add to the confusion for an improperly-trained professional.

Certain steps have been taken in this regard recently, with a spread of over-the-air testing. Proper procedure still has to be followed, but testing gets easier. As a test engineer you can have your test results processed and ready for use.

Analysis of test data through a computer by logging it directly into the system makes testing easier. Universal series bus (USB), LAN, general-purpose interface bus (GP-IB), RS232C are some of the remote connectivity options available with the latest equipment.

## No need of unnecessary measurements

Tripathi says, "Traditional power meters are broadband and have limited power ranges, so engineers and technicians are using spectrum analysers, which include many unneeded features, cost hundreds of thousands of dollars and take up half the test bench just to make simple, frequency based RF amplitude measurements." A long-standing trend has been to make the equipment and readings as easy as possible.

Nigam adds, "It is important that the chosen equipment has a bandwidth and sampling rate that is at least ten times the highest frequency to be measured so that high-frequency signals are not filtered out." This helps in high-frequency noise and harmonic content being taken into account while measuring root mean square values (RMS) and the actual heating effects of the signals.

Powermaster from Anritsu is a portable, USB-powered millimeter-wave power analyser, which enables frequency based measurement of RF power from 9kHz to 70GHz with capabilities as low as -90dBm. All this is available in a USB-powered device slightly bigger than a smartphone.

PA3000 from Tektronix allows for a one mega-cycle per second sampling rate and 1MHz bandwidth, thus, helping high bandwidth to combat most power applications. There is also the option of injecting an external signal as a source of frequency where measured signals are extremely distorted and direct frequency detection is challenging.